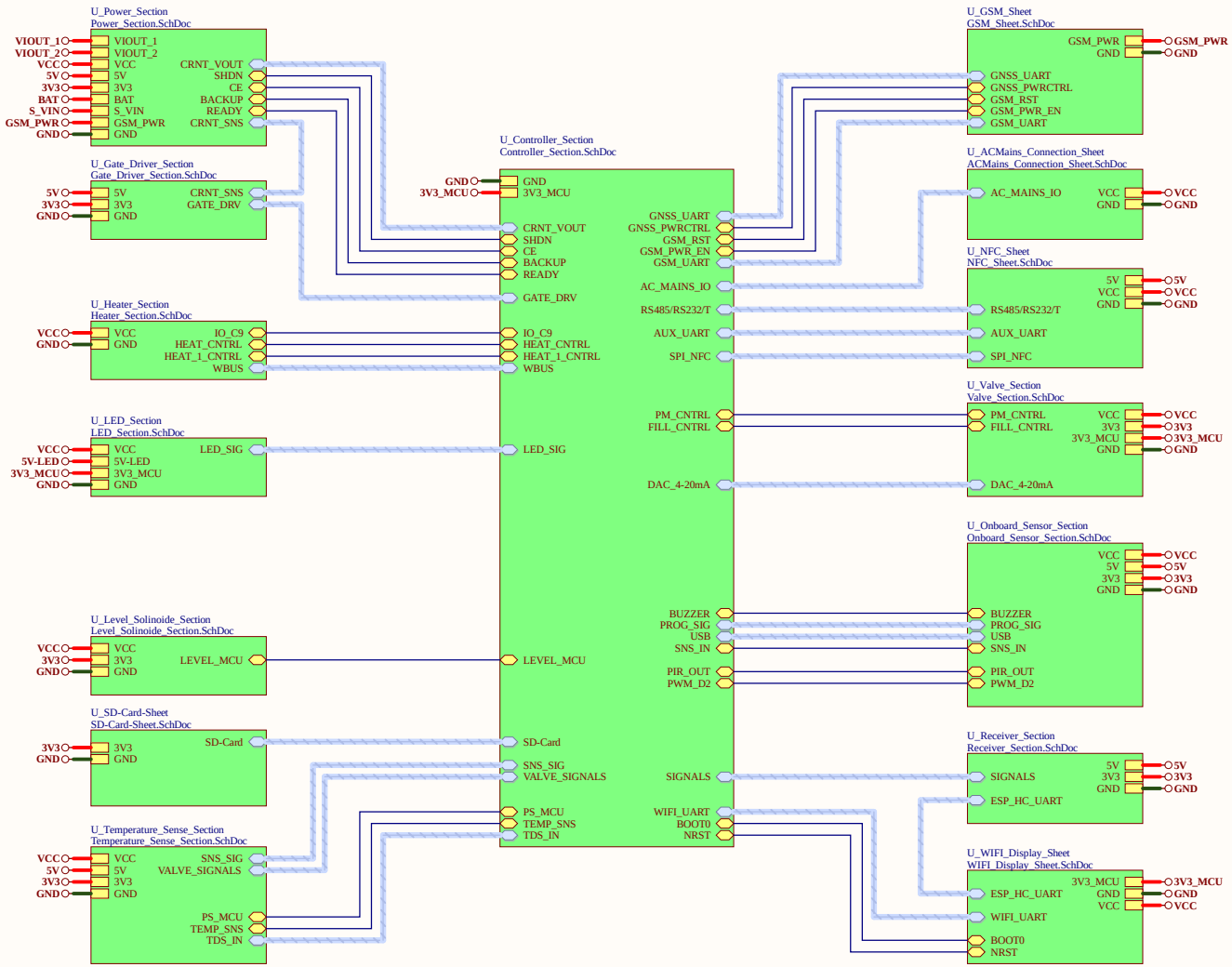
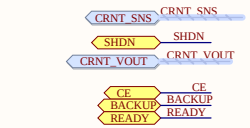
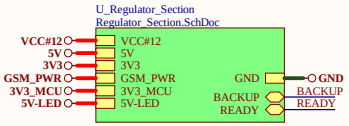
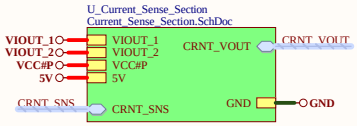


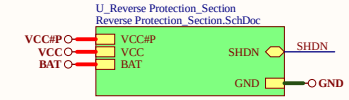
eSpark Consulting Group

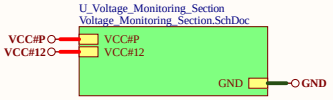
Title: Top_Sheet.SchDoc	
File: GTV4.0_Board.PrjPcb	
Date: 7/7/2025	Page Num: 1
Drawn By: Huzaifa Ali	Page Total: 17

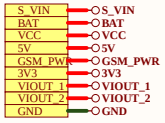


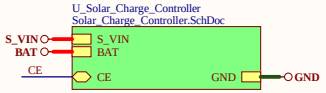








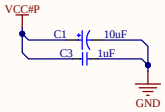




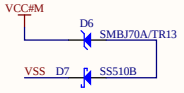
eSpark Consulting Group

Title: Power_Section.SchDoc		
File: GTV4.0_Board.PrjPcb		
Date: 7/4/2025	Page	Num: 2
Drawn By: Huzaifa Ali	Page	Total: 17

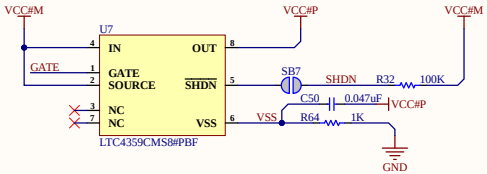
DECOUPLING CAPACITOR



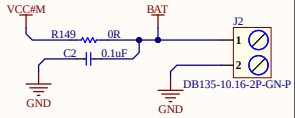
INPUT DIODES



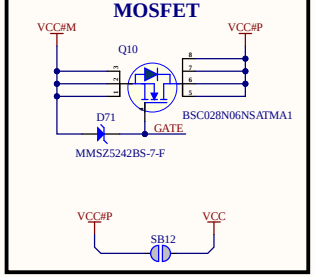
REVERSE POLARITY PROTECTION IC

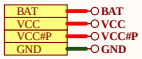


BATTERY / MAIN INPUT



MOSFET



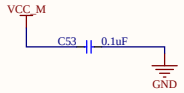




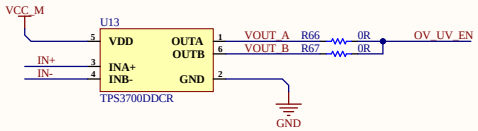
eSpark Consulting Group

Title: Reverse Protection_Section.SchDoc	
File: GTV4.0_Board.PrjPcb	
Date: 7/15/2025	Page Num: 3
Drawn By: Huzaifa Ali	Page Total: 17

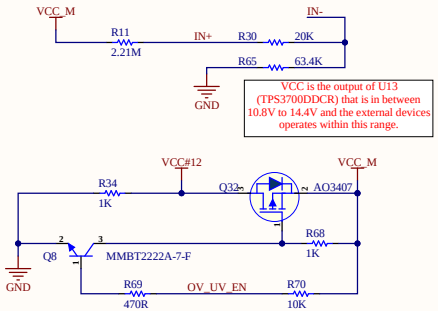
DECOUPLING CAPACITOR



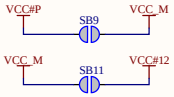
UNDER AND OVER VOLTAGE PROTECTION IC



SUPPORTING CIRCUITRY



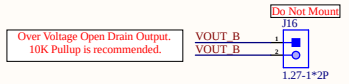
VOLTAGE TO MONITOR

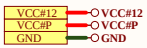


UNDER VOLTAGE OUTPUT



OVER VOLTAGE OUTPUT

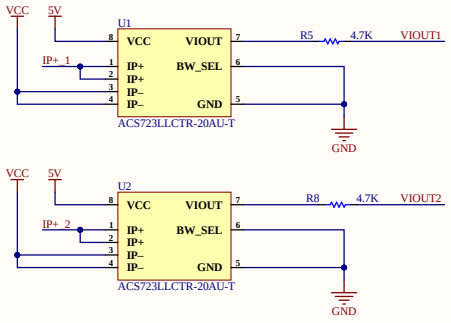




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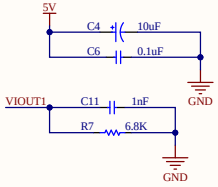
Title: Voltage_Monitoring_Section.SchDoc	
File: GTV4.0_Board.PrjPcb	
Date: 7/7/2025	Page Num: 4
Drawn By: Huzaifa Ali	Page Total: 17

CURRENT SENSE IC

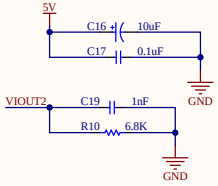


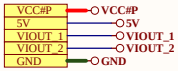
Two current sensing IC's are used to provide support to monitor current of two Pumps. Whereas currently we have only one pump in our V3 Controller.

VOUT1 SUPPORTING CIRCUIT



VOUT2 SUPPORTING CIRCUIT





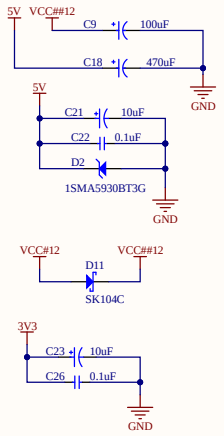




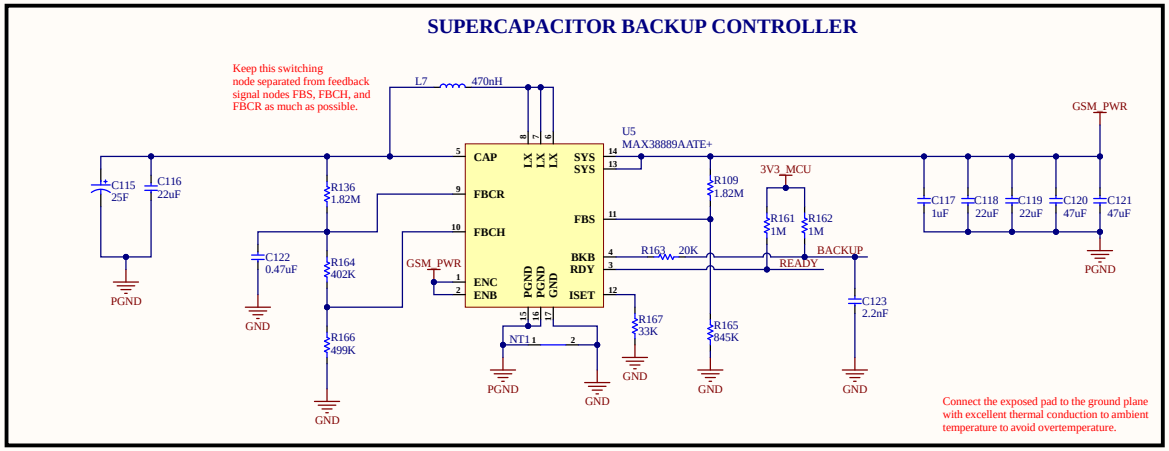
eSpark Consulting Group

Title: Current_Sense_Section.SchDoc	
File: GTV4.0_Board.PrjPcb	
Date: 7/7/2025	Page Num: 5
Drawn By: Huzaifa Ali	Page Total: 17

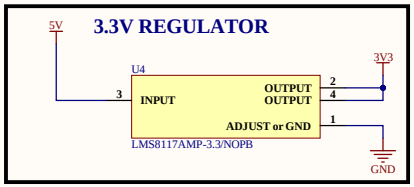
DECOUPLING CAPACITOR



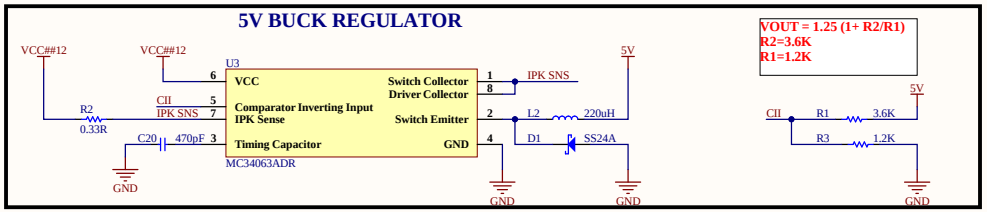
SUPERCAPACITOR BACKUP CONTROLLER



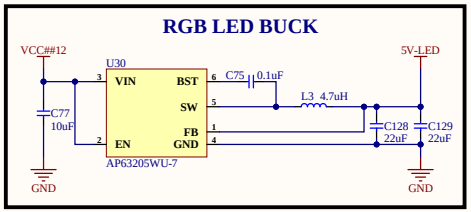
3.3V REGULATOR



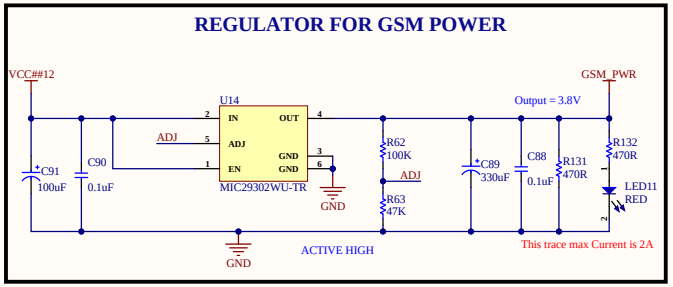
5V BUCK REGULATOR



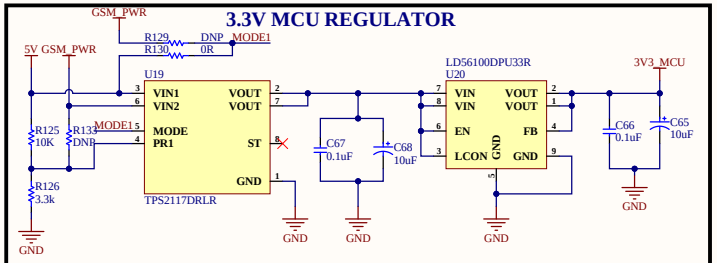
RGB LED BUCK

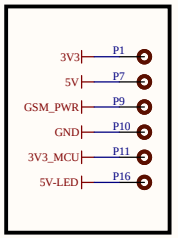


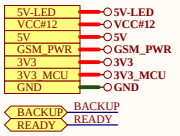
REGULATOR FOR GSM POWER



3.3V MCU REGULATOR



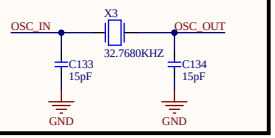




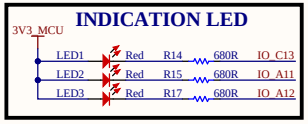
eSpark Consulting Group

Title: Regulator_Section.SchDoc	
File: GTV4.0_Board.PrjPcb	
Date: 7/17/2025	Page Num: 6
Drawn By: Huzaifa Ali	Page Total: 17

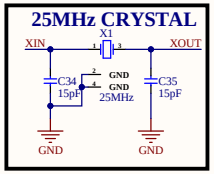
32.768KHZ CRYSTAL



INDICATION LED



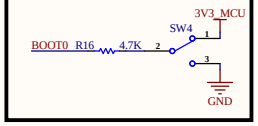
25MHz CRYSTAL



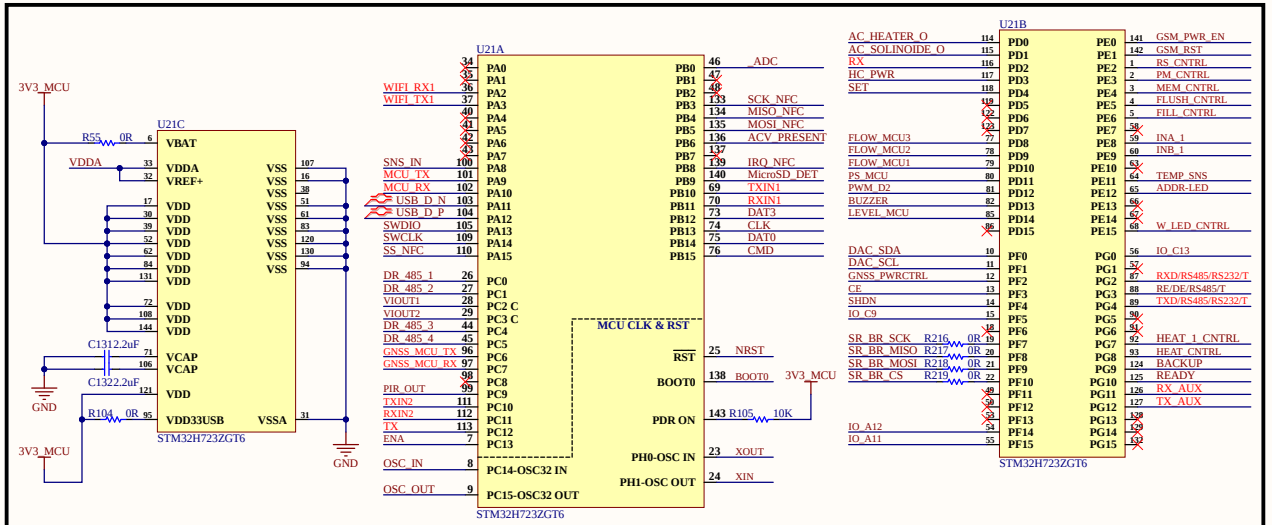
PULL UP



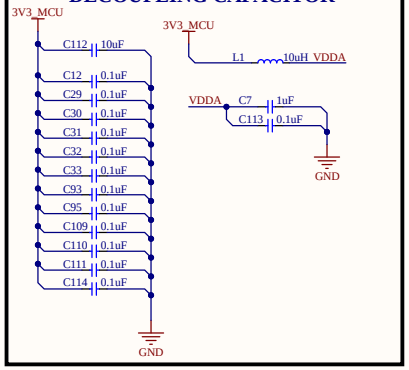
BOOT0 SWITCH



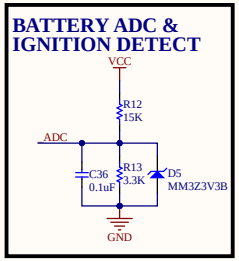
MAIN CONTROLLER



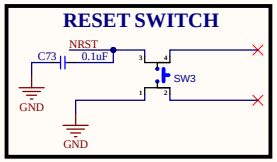
DECOUPLING CAPACITOR

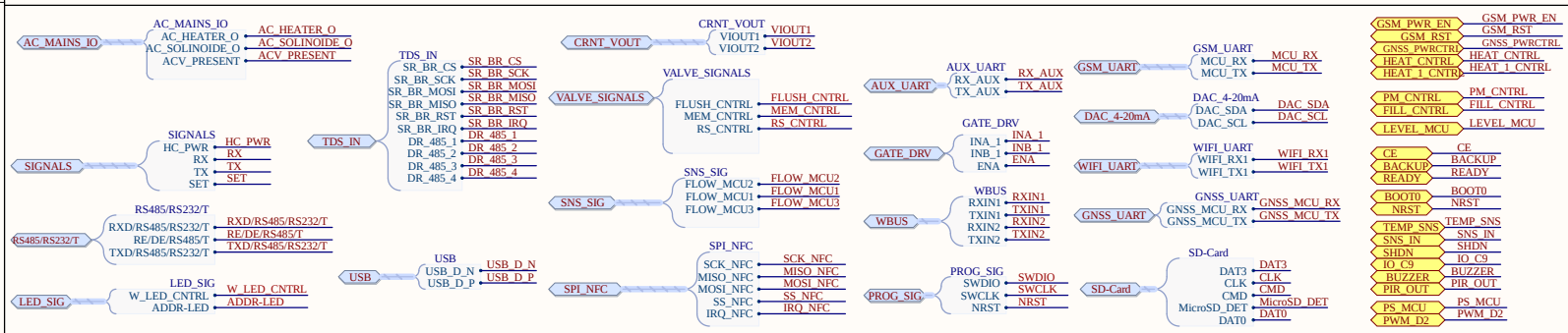


BATTERY ADC & IGNITION DETECT



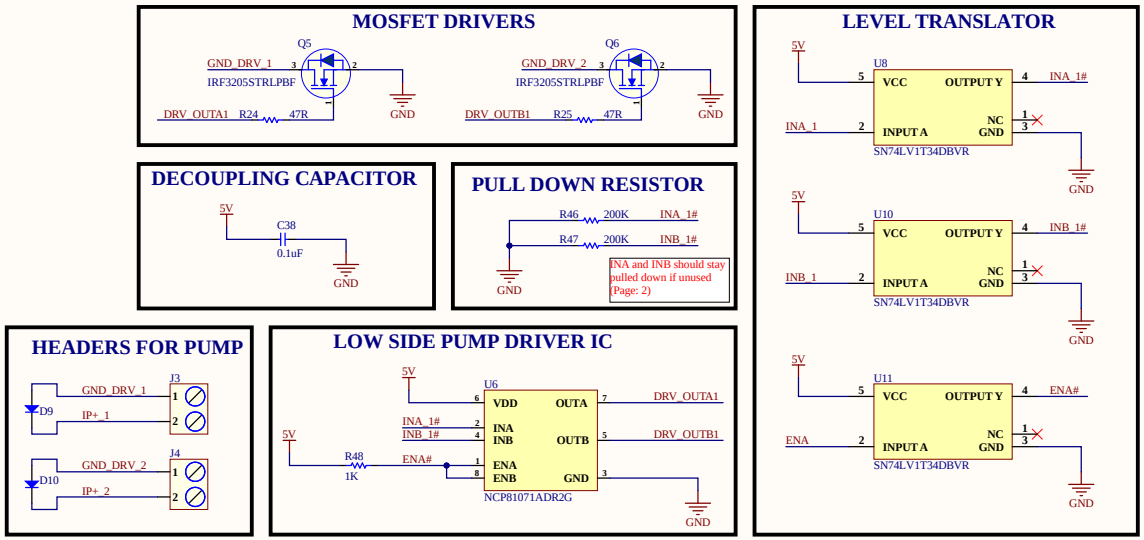
RESET SWITCH

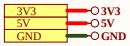




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Title: Controller_Section.SchDoc	
File: GTV4.0_Board.PrjPcb	
Date: 7/7/2025	Page Num: 7
Drawn By: Huzaifa Ali	Page Total: 17





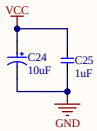




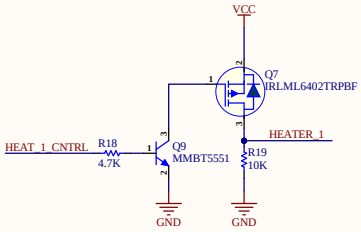
eSpark Consulting Group

Title: Gate_Driver_Section.SchDoc	
File: GTV4.0_Board.PrjPcb	
Date: 7/16/2025	Page Num: 10
Drawn By: Huzaifa Ali	Page Total: 17

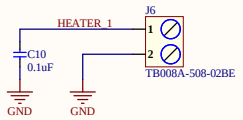
DECOUPLING CAPACITOR

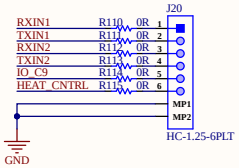


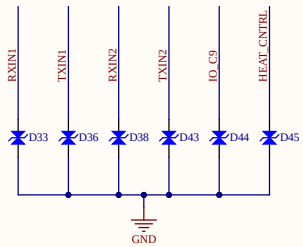
MOSFET CIRCUITRY



HEATER OUT







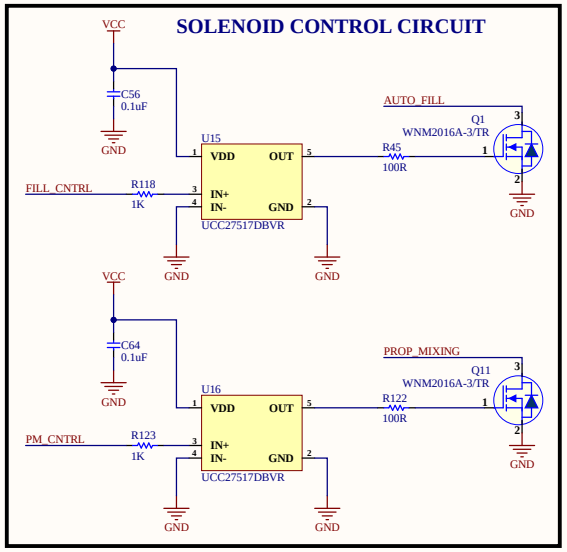


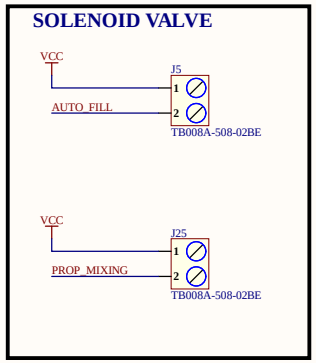


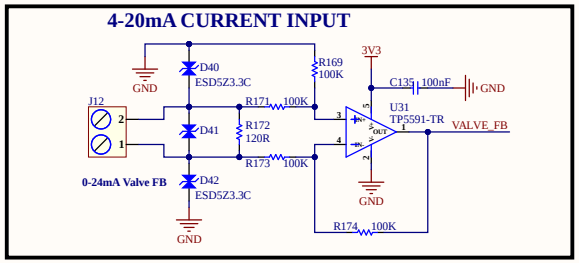


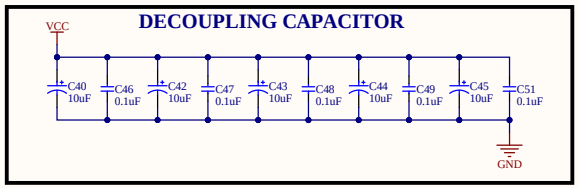
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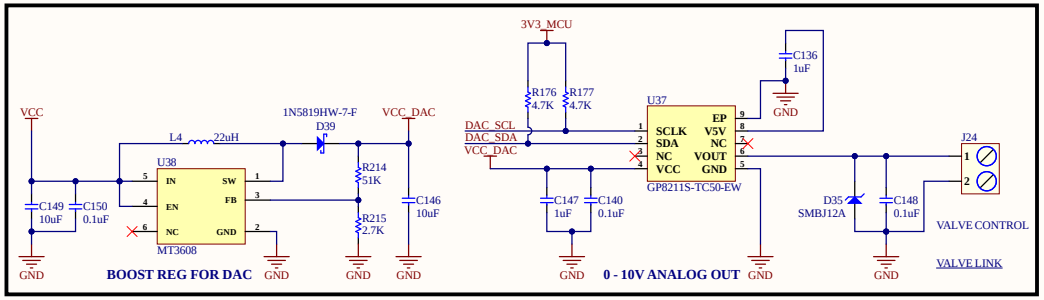
Title: Heater_Section.SchDoc		
File: GTV4.0_Board.PrjPcb		
Date: 7/16/2025	Page	Num: 11
Drawn By: Huzaifa Ali	Page	Total: 17











3V3_MCU — 3V3_MCU
3V3 — 3V3
VCC — VCC
GND — GND

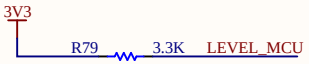
DAC_4-20mA — DAC_4-20mA
DAC_SDA — DAC_SDA
DAC_SCL — DAC_SCL

PM_CNTRL — PM_CNTRL
FILL_CNTRL — FILL_CNTRL

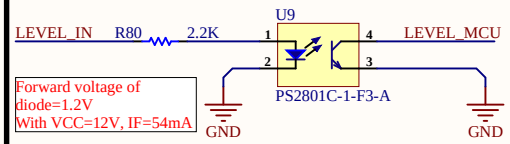
eSpark Consulting Group

Title: Valve_Section.SchDoc	
File: GTV4.0_Board.PrjPcb	
Date: 7/16/2025	Page Num: 12
Drawn By: Huzaifa Ali	Page Total: 17

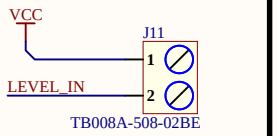
SUPPORTING CIRCUIT

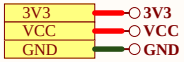


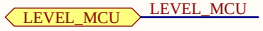
OPTO ISOLATION FOR LEVEL / FLOAT



LEVEL SENSOR INPUT



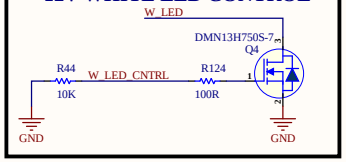




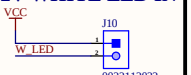
eSpark Consulting Group

Title: Level_Solinoide_Section.SchDoc	
File: GTV4.0_Board.PrjPcb	
Date: 7/16/2025	Page Num: 13
Drawn By: Huzaifa Ali	Page Total: 17

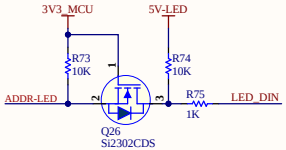
12V WHITE LED CONTROL



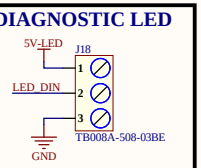
12V WHITE LED IN

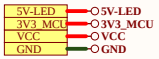


ADDRESSABLE LEDs LEVEL SHIFTER



DIAGNOSTIC LED



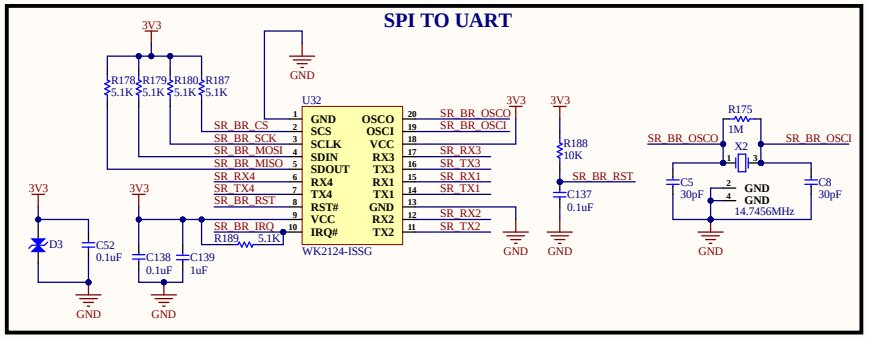


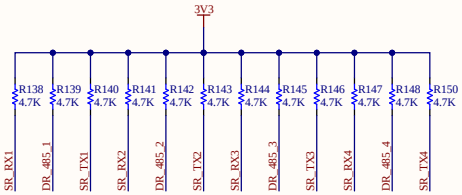


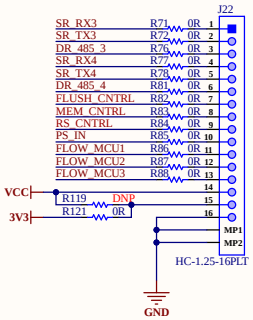
eSpark Consulting Group

Title: LED_Section.SchDoc		
File: GTV4.0_Board.PrjPcb		
Date: 7/16/2025	Page	Num: 14
Drawn By: Huzaifa Ali	Page	Total: 17

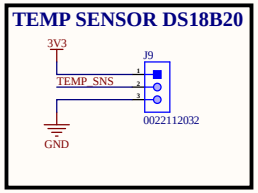
SPI TO UART



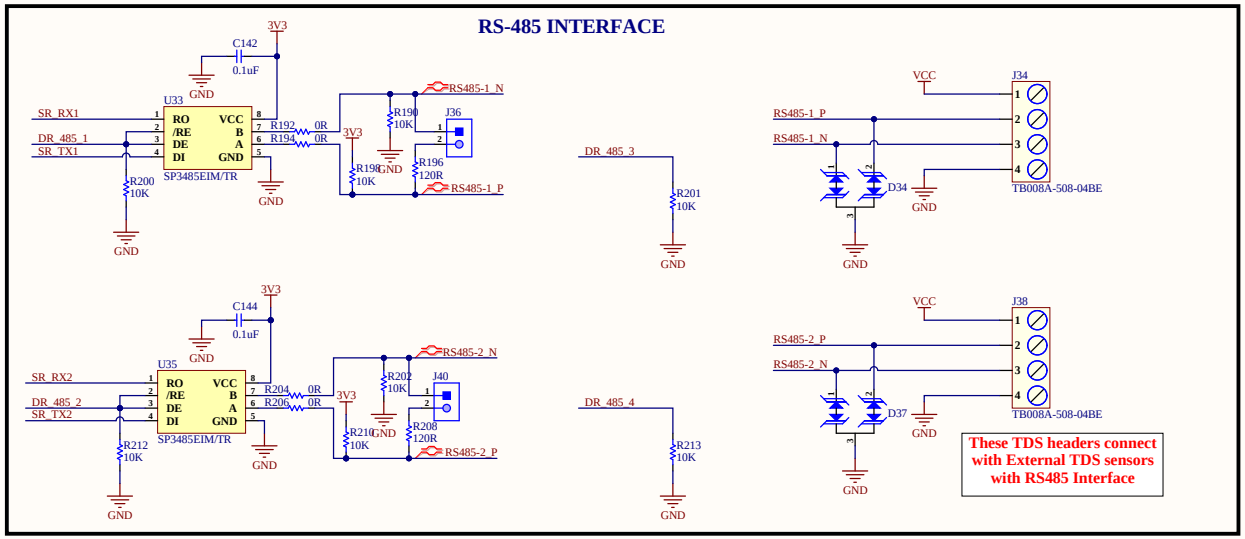




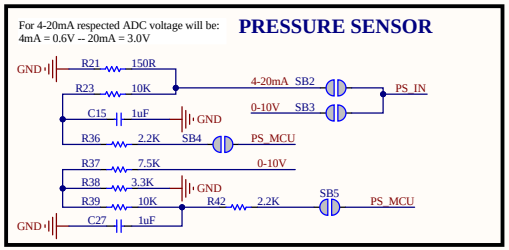
TEMP SENSOR DS18B20

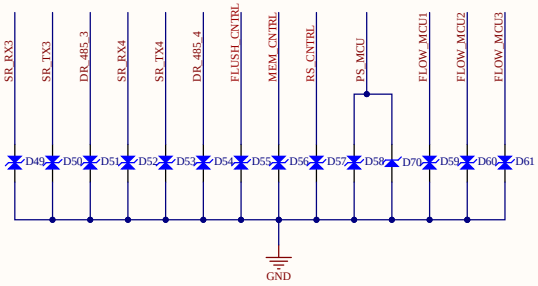


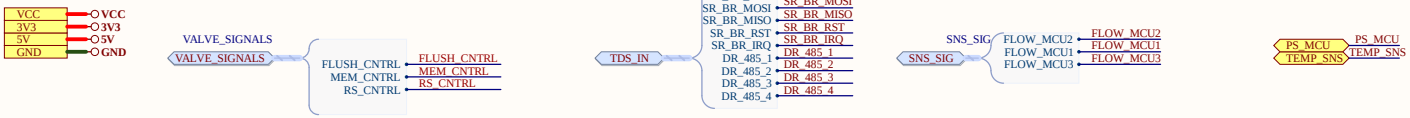
RS-485 INTERFACE



PRESSURE SENSOR



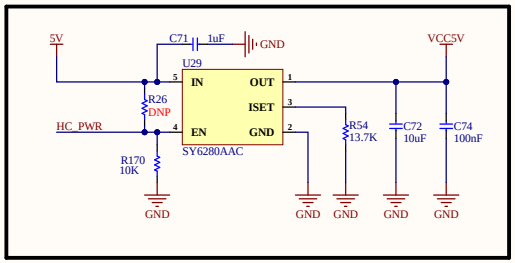




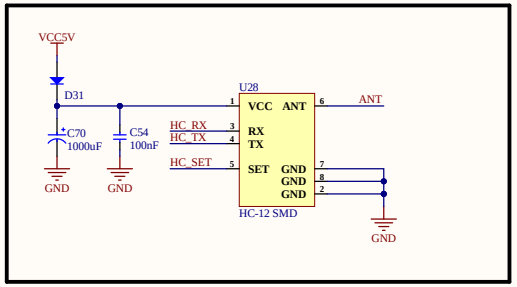
eSpark Consulting Group

Title: Temperature_Sense_Section.SchDoc		
File: GTV4.0_Board.PrjPcb		
Date: 7/17/2025	Page	Num: 15
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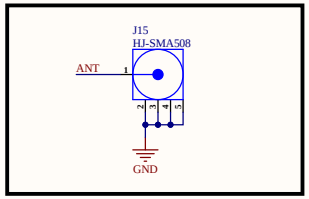
POWER SWITCH



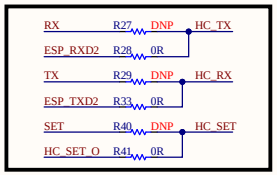
HC-12 433MHz MODULE

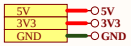


SMA CONNECTOR



CONTROL MCU SELECT





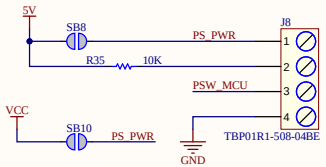




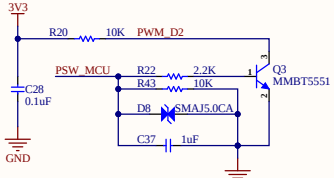
eSpark Consulting Group

Title: Receiver_Section.SchDoc		
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Date: 7/4/2025	Page	Num: 16
Drawn By: Huzaifa Ali	Page	Total: 17

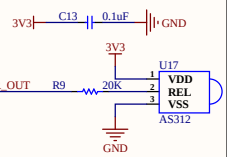
PRESSURE SWITCH



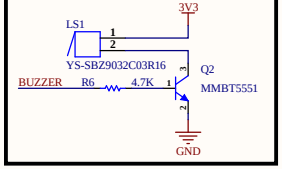
SUPPORTING CIRCUIT



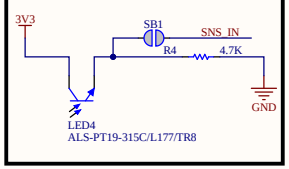
MOTION DETECTION



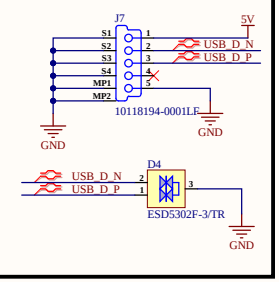
BUZZER CIRCUIT



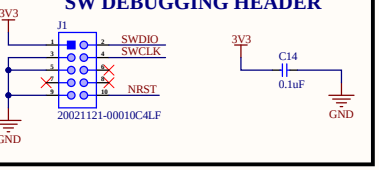
PHOTOTRANSISTOR

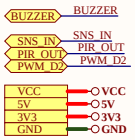


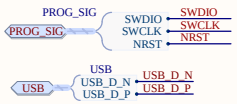
USB TYPE B CONNECTOR



SW DEBUGGING HEADER

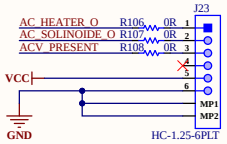


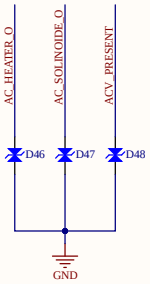


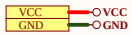


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Title: Onboard_Sensor_Section.SchDoc	
File: GTV4.0_Board.PrjPcb	
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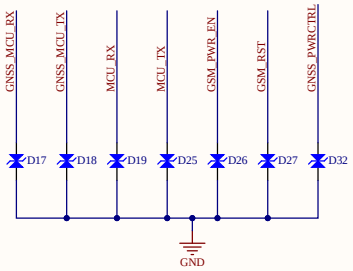


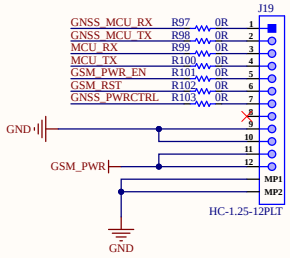




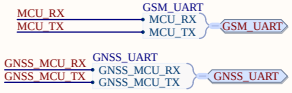
eSpark Consulting Group

Title: ACMains_Connection_Sheet.SchDoc	
File: GTV4.0_Board.PrjPcb	
Date: 7/7/2025	Page Num: *
Drawn By: Huzaifa Ali	Page Total: *





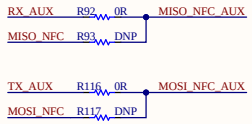


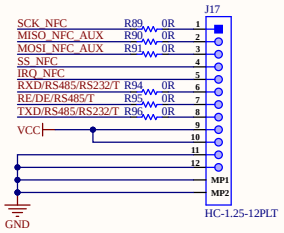


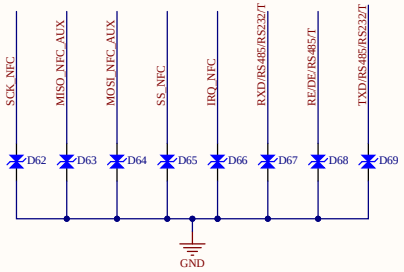


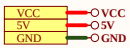
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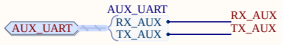
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Date: 7/7/2025	Page Num: *
Drawn By: Huzaifa Ali	Page Total: *



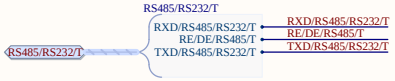






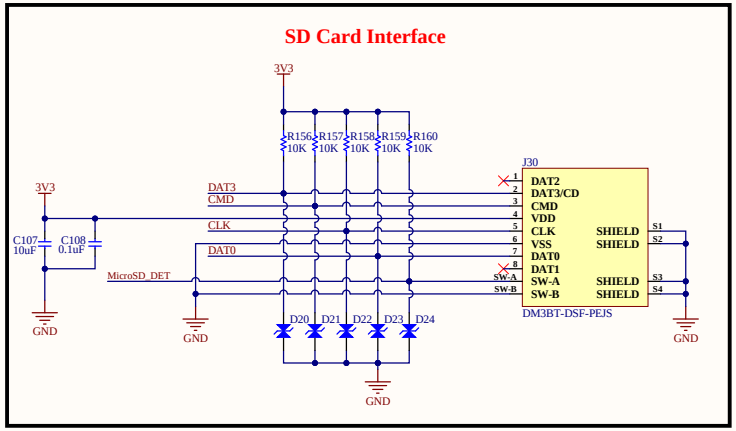






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Title: NFC_Sheet.SchDoc		
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Date: 7/16/2025	Page	Num: *
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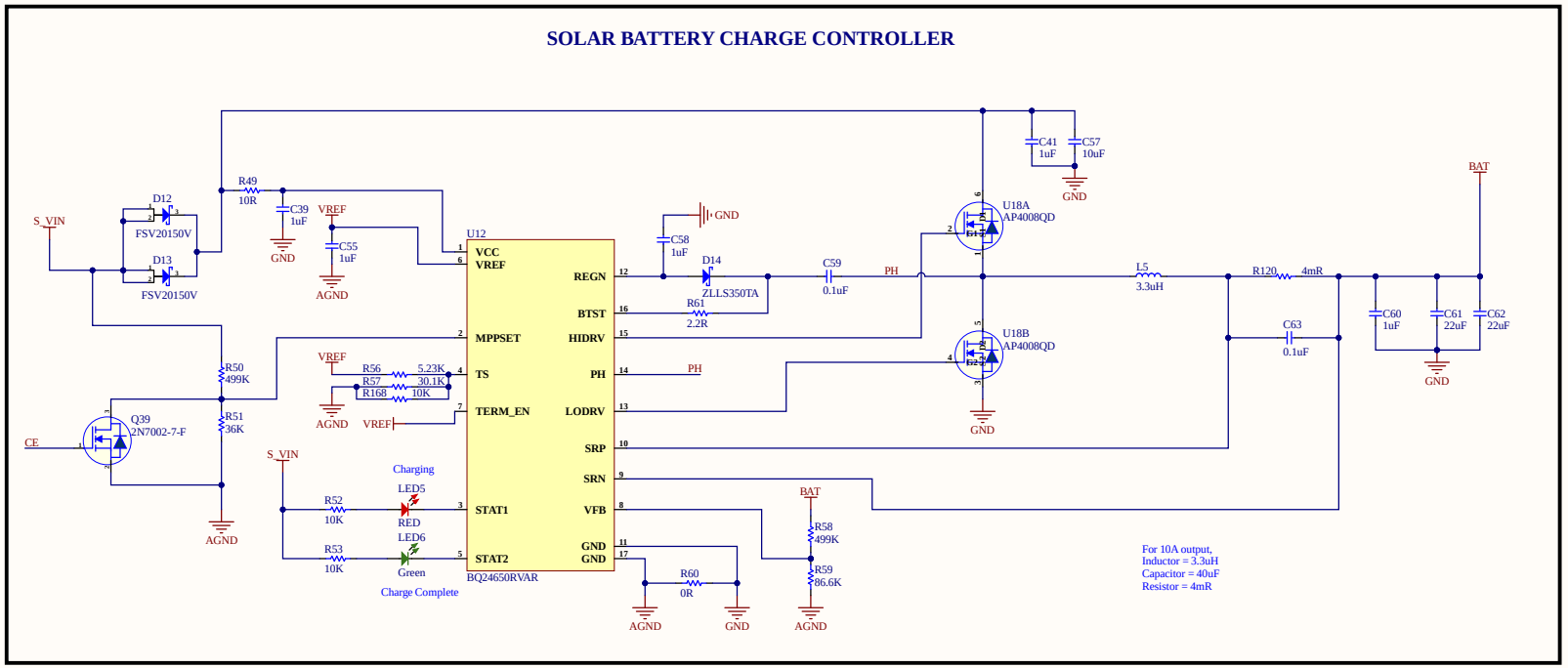
3V3
GND
3V3
GND

SD-Card
SD-Card
DAT3
CLK
CMD
MicroSD_DET
DAT0
DAT3
CLK
CMD
MicroSD_DET
DAT0

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Title: SD-Card-Sheet.SchDoc	
File: GTV4.0_Board.PrjPcb	
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Drawn By: Huzaifa Ali	Page Total: *

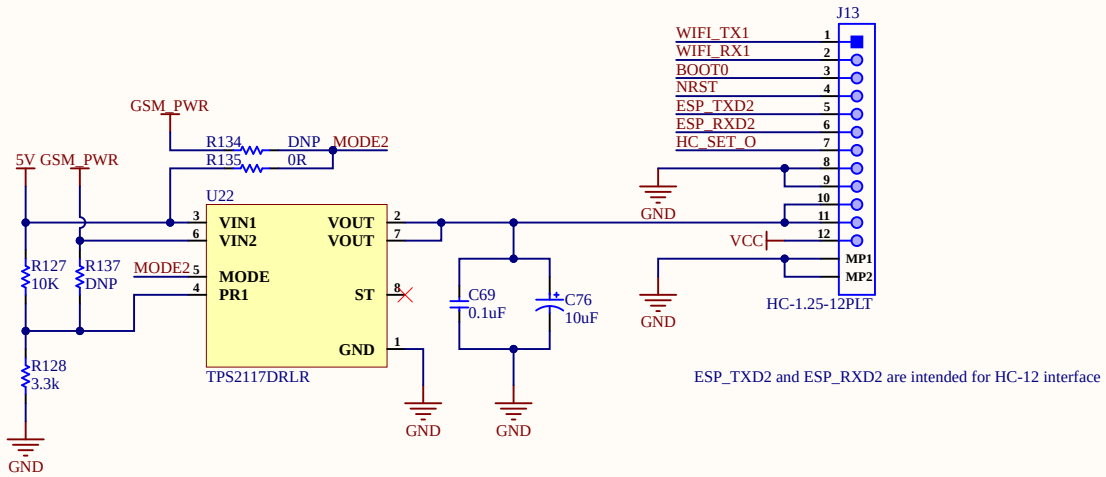
SOLAR BATTERY CHARGE CONTROLLER

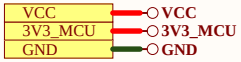


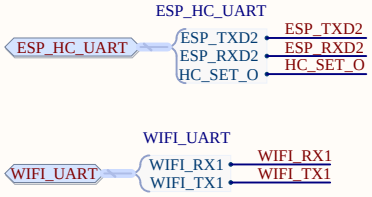
S_VIN → S_VIN
BAT → BAT
GND → GND
CE → CE

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Title:	Solar_Charge_Controller.SchDoc
File:	GTV4.0_Board.PrjPcb
Date: 7/16/2025	Page Num: *
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Title: WIFI_Display_Sheet.SchDoc	
File: GTV4.0_Board.PrjPcb	
Date: 7/17/2025	Page Num: *
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